

CHAPTER 8

The p-Block Elements

Group 13 Elements (Boron Family)

- The correct order of atomic radii in group 13 elements is (2018)
 - $B < Al < In < Ga < Tl$
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 - $B < Ga < Al < Tl < In$
- Which one of the following elements is unable to form MF_6^{3-} ion? (2018)
 - Ga
 - Al
 - In
 - B
- AlF_3 is soluble in HF only in presence of KF. It is due to the formation of: (2016-II)
 - $K_3[AlF_6]$
 - $K_3[AlF_4]$
 - AlH_3
 - $K[AlF_3H_3]$
- The stability of +1 oxidation state among Al, Ga, In and Tl increases in the sequence: (2015 Re)
 - $In < Tl < Ga < Al$
 - $Ga < In < Al < Tl$
 - $Al < Ga < In < Tl$
 - $Tl < In < Ga < Al$

Compounds of Boron and Uses of B and Al and Their Compounds

- Which of the following statement is not correct about diborane? (2022)
 - Both the Boron atoms are sp^2 hybridised.
 - There are two 3-centre-2-electron bonds.
 - The four terminal B-H bonds are two centre two electron bonds
 - The four terminal Hydrogen atoms and the two Boron atoms lie in one plane.
- Boric acid is an acid because its molecule: (2016 - II)
 - Accepts OH^- from water releasing proton
 - Combines with proton from water molecule
 - Contains replaceable H^+ ion
 - Gives up a proton

Group 14 Elements (Carbon Family)

- Match the following : (2020)

Oxide	Nature
A. CO	(i) Basic
B. BaO	(ii) Neutral
C. Al_2O_3	(iii) Acidic
D. Cl_2O_7	(iv) Amphoteric

 Which of the following is correct option?
 - (ii) (i) (iv) (iii)
 - (iii) (iv) (i) (ii)
 - (iv) (iii) (ii) (i)
 - (i) (ii) (iii) (iv)
- Which of the following oxide is amphoteric in nature? (2020-Covid)
 - SiO_2
 - GeO_2
 - CO_2
 - SnO_2
- Which of the following species is not stable? (2019)
 - $[SiF_6]^{2-}$
 - $[GeCl_6]^{2-}$
 - $[Sn(OH)_6]^{2-}$
 - $[SiCl_6]^{2-}$
- Which of the following is incorrect statement? (2019)
 - PbF_4 is covalent in nature
 - $SiCl_4$ is easily hydrolysed
 - GeX_4 ($X = F, Cl, Br, I$) is more stable than GeX_2
 - SnF_4 is ionic in nature
- It is because of inability of ns^2 electrons of the valence shell to participate in bonding that: (2017-Delhi)
 - Sn^{4+} is reducing while Pb^{4+} is oxidising
 - Sn^{2+} is reducing while Pb^{4+} is oxidising
 - Sn^{2+} is oxidising while Pb^{4+} is reducing
 - Sn^{2+} and Pb^{2+} are both oxidising and reducing

Allotropes of Carbon

- Choose the correct statement: (2022)
 - Both diamond and graphite are used as dry lubricants.
 - Diamond and graphite have two dimensional network
 - Diamond is covalent and graphite is ionic
 - Diamond is sp^3 hybridised and graphite is sp^2 hybridized.

13. Which of the following structure is similar to graphite? (2013)

- a. BN b. B
c. B_4C d. B_2H_6

Some Important Compounds of C & Si

14. Identify the correct statements from the following : (2020)

- (1) $CO_2(g)$ is used as refrigerant for ice-cream and frozen food.
(2) The structure of C_{60} contains twelve six carbon rings and twenty five carbon rings.
(3) ZSM-5, a type of zeolite, is used to convert alcohols into gasoline.
(4) CO is colorless and odourless gas.

- a. (1) and (3) only b. (2) and (3) only
c. (3) and (4) only d. (1), (2) and (3) only

15. Which of the following is not correct about carbon monoxide? (2020)

- a. It reduces oxygen carrying ability of blood.
b. The carboxyhaemoglobin (haemoglobin bound to CO) is less stable than oxyhaemoglobin.
c. It is produced due to incomplete combustion.
d. It forms carboxyhaemoglobin.

16. Which of these is not a monomer for a high molecular mass silicone polymer? (2013)

- a. $MeSiCl_3$ b. Me_2SiCl_2
c. Me_3SiCl d. $PhSiCl_3$

17. The basic structural unit of silicates is: (2013)

- a. SiO^- b. SiO_4^{4-}
c. SiO_3^{2-} d. SiO_4^{2-}

Answer Key

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17
c	d	b	c	a	a	a	d	d	a	b	d	a	c	b	c	b